

MTH1001 Mathematical Structures 2025–26, Term 1

Problem Sheet 1

This sheet is for discussion in the week 3 tutorials (Wednesday 8th October). Be ready to discuss (at least) Questions A5, A6(d), A7, B1(c)(d), B3 in the tutorial.

Your answers should use mathematical notation correctly and be written in logically correct sentences, including words such as "let", "if", "then". You should introduce any variables that occur in your argument.

SECTION A

These are fairly routine questions of a similar difficulty to those on tests and the mid-term exam.

A1. Consider the following sets:

$$A = \{3, 5, 6\}, \quad B = \{4, 5, 6, 7\}, \quad C = \{1, \{3, 5\}, \{3, 7\}\}.$$

Write each of the following sets as lists of elements enclosed in braces, **and** state the number of elements in each set. You need not provide your reasoning.

- (a) $A \cup B$;
- (b) $A \cap B$;
- (c) $(A \cap B) \cup C$;
- (d) $B \setminus A$;
- (e) $\mathcal{P}(A) \cap C$;
- (f) $(A \cap B) \times C$.

A2. Write each of the following sets as lists of elements enclosed in braces, **and** state the number of elements in each set. You need not provide your reasoning.

- (a) $\{n \in \mathbb{N} : n \text{ is a prime number and } n < 10\}$;
- (b) $\{n \in \mathbb{Z} : n^2 < 6\}$;
- (c) $\{x \in \mathbb{R} : 1 < x < 2 \text{ and } x^2 \in \mathbb{Z}\}$;
- (d) $\{x \in \mathbb{R} : |x| < 2 \text{ and } x^2 > 4\}$;
- (e) $\{n^2 : n \in \mathbb{Z} \text{ and } 2 \leq n \leq 3\}$;
- (f) $\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 < 4\}$;

(g) $\{\sqrt[3]{x} : x \in \mathbb{R} \text{ and } 1 < x < 2 \text{ and } x^2 \in \mathbb{Z}\};$

(h) $\{A \subseteq \{6, 7, 8\} : |A| = 2\}.$

A3. Let

$$A = \{m + n : m, n \in \mathbb{Z} \text{ with } m \text{ even, and } n \text{ odd}\},$$

$$B = \{k \in \mathbb{Z} : k \text{ is odd}\}.$$

Prove that $A = B$.

A4. Let

$$S = \{x \in \mathbb{R} : x \geq 0\} \text{ and } T = \{u^2 + v^2 : u, v \in \mathbb{R}\}.$$

Prove that $S \subseteq T$. Is it true that $S = T$? [You can assume any standard properties of real numbers.]

A5. Let

$$U = \{k + m + n : k, m, n \in \mathbb{N} \text{ with } k, m, n \text{ all odd}\}.$$

$$V = \{j \in \mathbb{N} : j \text{ is odd}\}.$$

Prove that $U \subseteq V$ but $V \not\subseteq U$.

A6. For each of the propositions (a)–(f) below:

- (i) Write the proposition out in English.
- (ii) State whether it is true or false. (No proof is required.)
- (iii) Write down the negation of the proposition, using universal and existential quantifiers ($\forall$ and $\exists$). Your answer should not include the logical negation ($\neg$) symbol.

(a) $\forall x \in \mathbb{R} : 2x \geq x;$

(b) $\exists n \in \mathbb{Z} : n^2 + n = 6;$

(c) $\forall x \in \mathbb{R} : \forall y \in \mathbb{R} : x + y = y + x;$

(d) $\exists x \in \mathbb{R} : \forall y \in \mathbb{R} : x + y = y + x;$

(e) $\forall m \in \mathbb{N} : \exists n \in \mathbb{N} : n > m;$

(f) $\exists m \in \mathbb{N} : \forall n \in \mathbb{N} : m \leq n.$

A7. Let S be the set of all students at the University of Exeter, and let P be the set of all people (living or dead).

Consider the following two statements:

(a) $\exists p \in P : \forall s \in S : p \text{ is the mother of } s;$

(b) $\forall s \in S : \exists p \in P : p$ is the mother of s .

Write each of these statements in words, expressing it as naturally in English as you can.

One of these statements is true, and one is false. Which is which?

A8. Let P, Q, R be propositions. Verify each of the following logical identities by writing out the appropriate truth tables:

- (a) $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$;
- (b) $(P \vee Q) \wedge R = (P \wedge R) \vee (Q \wedge R)$;
- (c) $P \wedge (P \Rightarrow Q) = Q \wedge P$.

SECTION B

These are more challenging questions. There may be questions on the January exam of similar difficulty to these.

B1. Determine whether each of the following statements is true or false. If it is true, prove it. If it is false, state and prove its negation.

- (a) For any sets A and B , we have $A \cap B \subseteq A \cup B$.
- (b) There exist sets A and B such that $(A \setminus B) \cup (B \setminus A) = A \cup B$.
- (c) For all sets A and B , we have $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$.
- (d) For all sets A and B , we have $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.
- (e) There are no sets A and B such that $A \subseteq B$ and $A \in B$.

B2. Given an infinite sequence of sets $A_1, A_2, A_3, \dots$, we can define their intersection and union by

$$\bigcap_{n=1}^{\infty} A_n = \{x : x \in A_n \text{ for all } n \geq 1\}$$

and

$$\bigcup_{n=1}^{\infty} A_n = \{x : x \in A_n \text{ for some } n \geq 1\}.$$

Let

$$A_n = \left\{ x \in \mathbb{R} : -\frac{1}{n} < x < \frac{1}{n} \right\} \text{ for } n \in \mathbb{N}.$$

Prove that

$$\bigcap_{n=1}^{\infty} A_n = \{0\}$$

and find $\bigcup_{n=1}^{\infty} A_n$.

B3. Which of the following propositions can be deduced from the premises $P \vee Q$ and $\neg P$?

$P, Q, P \wedge Q, P \Rightarrow Q, Q \Rightarrow P$.

B4. Which of the following propositions can be deduced from the premises $P \Rightarrow Q$ and Q ?

$P, Q, \neg P, \neg Q, P \wedge Q, Q \vee P, Q \Rightarrow P, (\neg Q) \Rightarrow P$.

B5. In lectures we gave the classical proof that $\sqrt{2}$ is irrational. (This depends on the fact that if $n \in \mathbb{N}$ and n^2 is even, then n must be even.) Try to adapt this proof to show that

(a) $\sqrt[3]{2}$ is irrational;

(b) $\sqrt{3}$ is irrational.

What facts about the natural numbers are you using in your proofs?